# Jordan Firth

Lieutenant Colonel, USSF

## EDUCATION

2016–2019 PhD, Aerospace Engineering, North Carolina State University, Raleigh NC
Performed Zero-G parabolic test flights of novel deployment mechanisms for rollable spacecraft booms with funding from a NASA grant. Led a 14-member team to design, develop, and manufacture instrumentation, support equipment, and boom deployment mechanisms, including satisfying FAA airworthiness standards.

2009–2011 MS, Astronautical Engineering, USAF Institute of Technology, Wright-Patterson AFB OH
Focus on astrodynamics and spacecraft system design. Team lead for a two-semester student project building and testing a cubesat-sized academic spacecraft. Thesis research focused on spacecraft flywheel vibration.

2002–2006 BS, Engineering Mechanics, USAF Academy CO

## MILITARY EDUCATION

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| Present | Air War College, currently enrolled in distance learning |
| 2017 | Air Command and Staff College, completed via distance learning |
| 2012 | Squadron Officer School, completed in residence |

## ASSIGNMENTS

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| 2022–present | Assistant Professor, Department of Astronautics, USAF Academy CO |
| Jan–May 2025 | Acting Cadet Squadron Commander, Cadet Group 4, USAF Academy CO |
| 2019–2022 | Warfare Domains Branch Chief, DIA Def Tech & Long-Range Analysis Office, Charlottesville VA |
| 2016–2019 | PhD Student, North Carolina State University, Raleigh NC |
| 2014–2016 | Assistant Professor, Department of Astronautics, USAF Academy CO |
| 2011–2014 | DSP Ops Officer, SBIRS Sys Test Engineer, Infrared Space Systems Directorate, Los Angeles CA |
| 2009–2011 | Master’s Student, USAF Institute of Technology, Wright-Patterson AFB OH |
| 2008–2009 | Rocket Sled Test Engineer, 846 Test Squadron, Holloman AFB NM |
| 2006–2008 | Flight Test Engineer, 586 Flight Test Squadron, Holloman AFB NM |

## LEADERSHIP AND COMMISSIONING EDUCATION

2025 Jan–2025 May Acting Squadron Commander, CS-32/CS-37, Cadet Group 4, USAF Academy CO
Led 2 NCOs and 100 cadets while two commanders were on extended leave. Responsible for order and discipline. Mentored cadet squadron leadership teams. Taught commissioning education lessons. Advised group leadership and served as a bridge between CW and DF. Directly contributed to Wing policies, including AFCWI development. Led squadron to #1/40 finish in 1st ever Superintendent’s Challenge.

2019–2022 Chief, Warfare Domains Branch, DIA Def Tech & Long-Range Analysis Office, Charlottesville VA
Branch chief for 29 acquisition intelligence analysts at seven locations. Assessed near-term threats to US systems. Maintained the authoritative Defense Intelligence Threat Library, which connected program staff to intelligence authors for $1.23T worth of defense programs. As military advisor for a 150-person office, I also guided development of eleven assigned joint officers

2018 Jan–2018 May AS 222 Course Director, AFROTC Detachment 595, NC State University, Raleigh NC
Taught spring semester sophomore-level ROTC Team & Leadership Fundamentals class to 27 cadets at the Det commander’s request. Prepared cadets to put leadership concepts into practice at field training.

## ENGINEERING EDUCATION EXPERIENCE

2022–present Assistant Professor, Department of Astronautics, USAF Academy CO

*Course Director, Astro 331 Space Systems Engineering*
Directed intermediate lab-based space systems engineering course teaching fluency in system-level design issues and all major spacecraft subsystems through assembly and operation of a breadboard-based spacecraft analog. Transitioned lab curriculum to use sustainable commodity hardware.

*Course Director, Astro 201 Tech. Skills for Astro/Engr 495 Tech. Skills & Domain Comm. for Engineers*
Developed course introducing key skills and tools for Astro majors, and incorporated instruction in engineering communication to address an externally-identified educational shortfall in the major. Course is on track for Fall 2026 inclusion in USAFA COI as a requirement for two engineering majors.

*Instructor, Astro 310 Engineering Principles of Space Warfighting*
Assisted Department-wide update of curriculum to include space warfighting topics in course required for all USAFA graduates.

*Director of Assessments, 2022–2024*
Maintained department assessment program supporting continuous improvement and periodic ABET accreditation. Led 2024 Engineering Program Advisory Council. Performed mid-cycle ABET self-assessment. Identified and filled gaps in department assessment program.

2014–2016 Assistant Professor, Department of Astronautics, USAF Academy CO

*Course Director, Astro 321: Intermediate Astrodynamics*
Instructed intermediate course in orbital mechanics including orbit determination, maneuvers, perturbations, and rendezvous and proximity operations, as well as programming skills to perform these operations at scale.

*Course Director, Astro 310: Introduction to Astronautical Engineering*
Directed introductory astrodynamics course, including orbital mechanics, spacecraft systems, and rocket propulsion. Required for all Air Force Academy cadets. Trained incoming faculty before their first teaching semester.

*Instructor, EM 320: Dynamics*
Instructed introductory dynamics, covering kinematic and kinetic analysis of particles and rigid bodies, as well as an introduction to mechanical vibrations of simple systems. Required for all Aeronautical, Astronautical, and Mechanical Engineering Majors.

*Engineering Division Executive Officer, 2013–2014*

## OTHER PROFESSIONAL EXPERIENCE

2012–2014 SBIRS System Test Engineer, Infrared Space Systems Directorate, Los Angeles CA
Oversaw system test for SBIRS missile warning system, including pre- and post-launch checkouts for three spacecraft. Testing led to successful completion of a major in-place ground system upgrade with no downtime. Compiled and edited biennial update to the Congressionally-mandated Test and Evaluation Master Plan, including coordination with external agencies for OSD approval.

2011–2012 DSP Orbital Operations Officer, Infrared Space Systems Directorate, Los Angeles CA
Oversaw execution & renegotiation of ongoing factory support contract providing manufacturer expertise for DSP missile warning operators, enabling continuous global missile warning. Contractor activities included training all incoming personnel, anomaly resolution, and spacecraft health monitoring.

2008–2009 Rocket Sled Test Engineer, 846 Test Squadron, Holloman AFB NM
Analyzed rocket sled test profiles at Holloman High Speed Test Track. Responsible for sled structural margins, acceleration profiles, and rocket staging. Coordinated with program managers, in-house shops, track operators. Revived dormant squadron capability to fire captive Stinger missiles, providing accurate threat profiles for final operational checkout of CH-47 countermeasure system before fleet-wide deployment.

2006–2008 Flight Test Engineer, 586 Flight Test Squadron, Holloman AFB NM
Performed advanced avionics flight test and test support using highly-modified T-38B and C-12J aircraft. These aircraft were used to carry test equipment and serve as targets for urgent DOD priority missions. Planned photo/safety chase for the final pre-deployment test of JASSM-ER, which verified safe missile separation and a successful 500+ mile flight. Coordinated final pre-delivery test of Iraq's first air-to-ground missile capability—Hellfire missiles launched from a Cessna Caravan.

## PUBLICATIONS

Peer-Reviewed Journals

Firth, Jordan A., and Mark R. Pankow. "Minimal unpowered strain-energy deployment mechanism for rollable spacecraft booms: ground test." *Journal of Spacecraft and Rockets*. 2020

Firth, Jordan A., and Mark R. Pankow. "Advanced dual-pull mechanism for deployable spacecraft booms." *Journal of Spacecraft and Rockets*. 2019

Brown, Robert B., and Jordan A. Firth. "Analysis of trans-Neptunian objects and a proposed theory to explain their origin." *Monthly Notices of the Royal Astronomical Society*. 2016

Firth, Jordan, and Jonathan Black. "Vibration interaction in a multiple flywheel system." *Journal of Sound and Vibration*. 2012

White, Charles, Jordan Firth, and Mark Pankow. "Impact of Joint Parameters on Performance of Self-Opening Dual-Matrix Composites." *Journal of Spacecraft and Rockets.* 2023

Conference Papers

Adamcik, Benjamin, Jordan Firth, et al. "Impact of storage time and operational temperature on deployable composite booms." *AIAA Scitech 2020 Forum*. 2020

Firth, Jordan, et al. "Minimal Unpowered Strain-Energy Deployment Mechanism for Rollable Spacecraft Booms." *AIAA Scitech 2019 Forum*. 2019

Supported Cadet Research

Kison, Sydney, Jeremy Chen, Sonja Nelson, and Austin Smith. "Small Satellite Reaction Wheel Manufacturing." *Unpublished, presented at 2024 AIAA Region V student conference. 2024*

Burkhart, Devin M., John R. Snider, and Jordan A. Firth. "Flight Qualification for Low Cost Student Manufactured and Balanced Reaction Wheel." *Unpublished, presented at 2023 AIAA Region V student conference*. 2023

Other Publications

Firth, Jordan "Drag Coefficient of a Model Rocket Recovery Parachute." *Peak of Flight Newsletter*. Apogee Components Inc. 2025

## AWARDS

2024 USAFA/DFAS Wittry Award for Engineering Design
2022 White House Fellows Program Regional Finalist